# Patrick Mockridge

Aberystwyth, Wales · patrickmockridge@gmail.com · patrickmockridge.github.io

## PROFILE AND METHOD

Process safety taught me that a plausible number is not a safe number — so every result here is either proved or carries its limits. I was a process safety engineer from 2006 to 2015, chartered in 2013, and have worked independently since 2016 on protocol software, formal verification, and numerical systems. The method is the same in both fields: specify the failure mode before the architecture; prove what can be proved; contain what cannot; record provenance with every result. Where proof stops is stated in the limits.

The three projects are three instances of that method. DarkWow is containment: private coordination on a public chain, with authority held as an object capability rather than a public administrator. rchain-rust is rescue: a dissolved network's node rebuilt in Rust, with its language laws machine-checked at the model level and conformance-tested at the node. Azoth is provenance: chemical-engineering calculations where every number carries its source and verification state, and unsupported answers are refused.

## SELECTED WORK

### DarkWow

Designed for private organisations on a public chain, without an administrator or a public voter roll

2026– · Research prototype, unaudited

### rchain-rust

A rescued blockchain node whose language laws are machine-checked

2026 · Pre-release

### Azoth

A process calculation library where every number carries its provenance

2026 · v0.1.x, not for design work

Records: patrickmockridge.github.io/?tab=darkwow · /?tab=rchain-rust · /?tab=azoth

## CHRONOLOGY

| PERIOD | WORK |
|---|---|
| 2016–present | Independent work: protocol software, formal verification, numerical systems. RChain token originator; multi-signature DAO contract task force. Earlier advisory work includes Enjin, IEBC/CoEngineers, Synsapien, CubeSpawn, and ECP24. |
| 2011–2014 | Bitcoin since 2011; London Ethereum community, 2014. |
| 2006–2015 | Chemical engineer, process safety; UK and North Africa. Chartered Professional Chemical Engineer, 2013. |
| 2002–2006 | MEng Chemical Engineering, University of Edinburgh. |

## RESEARCH AND WRITING

UNISS / Ingenesist; Romeo–Juliet framework. Selected writing.